NATIONAL LOTTERY AUTHORITY • COMPLIANCE REFERENCE

LuckyEthio Raffle Platform

Comprehensive Technical System Architecture, Provably Fair Cryptographic Engine, Concurrency Queue, Omnichannel USSD/POS & Operational Manual.

Platform Release: Version 2.0 (Turborepo Monorepo Architecture)
Compliance Standard: SHA-256 Commit-Reveal RNG • Zero-Trust Deterministic Verification
Statutory Permit: NLA/ETH/2026/89 (National Lottery Administration)

Document Scope:
System Architecture, End-to-End Ticketing, 1-Hour Hold Engine, Payment Adapters & Verification.

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Web Customer Portal:
http://localhost:3000

Admin Operations Console:
http://localhost:3001

1. Executive Summary & Monorepo Layout

LuckyEthio is an enterprise-grade, provably fair lottery and raffle ticketing ecosystem built to serve both digital smartphone users and offline citizens across Ethiopia. The platform resolves the historical trust deficit in traditional lotteries through pre-committed cryptographic hashing while supporting high-concurrency ticket minting.

1.1 Monorepo Structure (Turborepo)

Workspace	Port / Path	Primary Responsibility
@raffle/web	Port 3000	Customer Web Portal, Raffle Catalog, 4-State Ticket Matrix, Payment Simulator, My Tickets QR Receipts, Agent POS Terminal, USSD Dial Simulator, Public Cryptographic Verifier.
@raffle/admin	Port 3001	Admin Operations Console, Real-time Sales Telemetry, Raffle Manager with SHA-256 Pre-commitments, Agent KYC & Float Approvals, Live Provably Fair Draw Console, Financial Ledger.
@raffle/database	packages/database	Prisma ORM schema definitions, SQLite / PostgreSQL data access layer, migrations, and automated database seeds.
@raffle/shared	packages/shared	Shared business logic: SHA-256 commit-reveal RNG, in-memory concurrency mutex queue, 1-hour ticket hold engine, multi-gateway payment adapters, USSD state machine, bilingual dictionaries.

2. Provably Fair SHA-256 Cryptographic Engine

To eliminate insider fraud, winner selection is governed by the **Commit-Reveal Cryptographic Scheme**. The winning ticket number is mathematically locked before any ticket is purchased and can be independently audited by any citizen or regulator.

> **Core Invariant:** The platform cannot alter the winning ticket number after sales begin, because changing the outcome would alter the SHA-256 hash, causing the public cryptographic verification to fail.

2.1 Step-by-Step Cryptographic Flow

1. **Pre-Commitment (At Raffle Creation):**
   A secure 32-byte cryptographic secret seed is generated. The system computes and publicly publishes:

   ```
   CommitHash = SHA256(SecretSeed + ":" + RaffleID + ":" + TotalTickets)
   ```

2. **Ticket Sales (Zero Knowledge):**
   Customers purchase tickets (e.g. 1 to 1,000). The plaintext `SecretSeed` remains encrypted and secret in the database.
3. **Live Draw Execution:**
   When all tickets are sold, the supervisor triggers the draw in the Admin Console. The system reveals the `SecretSeed` and executes the deterministic winning formula:

   ```
   WinningTicketNumber = (BigInt(SHA256(SecretSeed)) % TotalSoldTickets) + 1
   ```

4. **Independent Public Verification (/verifier):**
   Any customer or auditor pastes the revealed seed and commit hash into the Independent Verifier. The tool recomputes the SHA-256 hash and validates that the winner matches with 100% precision.

3. Concurrency & 1-Hour Booking Hold Engine

3.1 High-Concurrency Mutex Queue

Under flash sales (e.g. hundreds of users rushing to purchase lucky numbers like #7 or #77), traditional databases suffer from race collisions or duplicate ticket minting. LuckyEthio implements an in-memory sequential promise queue (`ConcurrencyQueue`) per raffle. Every reservation and purchase executes inside an isolated database transaction, guaranteeing **zero duplicate tickets** and **zero overselling**.

3.2 1-Hour Exclusive Reservation Engine

When a customer selects lucky numbers or triggers a quick-pick and proceeds to checkout, the system places a temporary **1-Hour Reservation Hold**:

- **Temporary Lock:** Tickets are marked as `RESERVED` with `reservedUntil = now + 60 minutes`.
- **Exclusivity:** Other buyers see these numbers as **Booked (1hr)** with a pulsing hourglass icon and cannot select them.
- **Live Countdown:** The payment drawer and checkout pages display an animated countdown timer (⏳ `59:45 remaining`).
- **Settlement:** If payment succeeds within 60 minutes, status converts to `CONFIRMED`. If 60 minutes elapse unpaid, `cleanupExpiredReservations` automatically deletes the hold, instantly returning the numbers to the available pool.

3.3 4-State Visual Ticket Matrix

State	Visual Color	Badge & Icon	Behavior & Action
Available	White Card (Slate border)	⚪ Available	Clickable, selectable by customer, green hover outline.
Selected	Vibrant Emerald Green	🟢 Selected ✓	Active in user's current checkout cart.
Booked	Warm Amber Yellow	🟡 Booked ⏳	Locked by another in-flight buyer for 60 min. Disabled.
Sold	Muted Gray / Strikethrough	⛔ ~~Sold 🔒~~	Permanently purchased & confirmed. Disabled.

4. Payment Gateways & Kiosk Agent POS

4.1 Ethiopian Payment Integrations

Gateway	Protocol	Target User Group
Telebirr	USSD Push / QR Code	Ethio Telecom mobile money subscribers across all regions.
CBE Birr	Mobile Banking API	Commercial Bank of Ethiopia account holders.
Chapa	Card & Bank Webhook	Multi-bank transfers, Visa, Mastercard, and international cards.
SantimPay	Direct Bank Transfer	Local Ethiopian commercial bank direct settlement.
Agent Cash	Kiosk POS Float	Cash-paying walk-in retail customers at physical stores.

4.2 Authorized Kiosk Agent POS & Double-Entry Float Ledger

Physical lottery kiosks operate via `/agent`. Agents maintain a prepaid float balance (e.g. 10,000 ETB). When selling tickets to cash-paying walk-in customers:

- Ticket price is atomically deducted from the agent's `floatBalance`.
- Agent's commission (5% to 10%) is instantly accrued and credited.
- Every transaction creates an immutable `AgentLedger` audit record with balance-after snapshots.

4.3 Offline GSM USSD State Machine (*804#)

Accessible via `/agent/ussd-simulator` or real GSM USSD gateway. Operates on 2G/3G basic feature phones with 0 KB mobile data usage. Supports active raffle browsing, Telebirr ticket purchases, checking purchased ticket numbers, and Agent POS cash sales.

5. Post-Draw Experience & 24-Hour Hero Winner Spotlight

5.1 Automated Pre-Draw & Winner Notifications

- **Pre-Draw Alert:** When a raffle reaches 100% sold-out tickets, all ticket buyers receive an automated in-app alert and SMS: *"All tickets sold! The Provably Fair live draw is commencing. Check your ticket numbers in My Tickets."*
- **Winner Announcement:** Following draw execution, all buyers receive the winning ticket announcement with the cryptographic proof. The winner receives a dedicated golden celebration alert.
- **Notification Bell:** The top navigation includes an interactive bell with unread counter badges.

5.2 24-Hour Hero Winner Spotlight Banner

Following draw execution, a celebratory full-width card is displayed at the top of the homepage (`/`) for **24 hours**:

- **Festive Confetti Shower:** Triggers automatically on visitor arrival.
- **Winner Credentials:** Displays the winner's name and masked mobile number (e.g. `Helen T. (+251933***566)`).
- **Winning Ticket # & Verification Code:** Highlighted in bold green badge (e.g. `Ticket #427`).
- **Grand Prize Display:** High-resolution prize photo and ETB market valuation.
- **24-Hour Remaining Timer:** Real-time timer showing remaining spotlight hours and minutes.
- **1-Click Mathematical Proof:** Direct button opening `/verifier` with pre-filled SHA-256 parameters.

5.3 Permanent Historical Archives

- **Winners Hall (/winners):** Permanent public record of all past completed draws, winning tickets, and NLA compliance audit records.
- **Customer History (/my-tickets):** Personal purchase history with digital QR receipts and winner ribbons.

6. Operational Runbook & Verification Suite

6.1 Pre-Configured Demo Accounts

Role	Full Name	Mobile Number	Portal Access
Super Admin	Abebe Kebede	+251911000000	Admin Console (:3001)
Lottery Admin	Sara Haile	+251911000001	Admin Console (:3001)
Kiosk Agent	Dawit Tadesse	+251912345678	Web POS / USSD (:3000)
Customer 1	Helen Tesfaye	+251933445566	Web Client (:3000)
Customer 2	Yohannes Girma	+251944556677	Web Client (:3000)

6.2 Automated Platform Test Suite

```
$ ./node_modules/.bin/tsx scripts/test-platform.ts ▶ TEST 1: Concurrency & Anti-Collision Ticket Minting -> PASSED (Zero duplicates, Zero oversell) ▶ TEST 2: Provably Fair SHA-256 Commit-Reveal Cryptography -> PASSED (100% mathematically sound) ▶ TEST 3: Agent POS Float Deductions & Auto-Commission -> PASSED (Float & Ledger verified) ▶ TEST 4: USSD Offline State Machine Engine (*804#) -> PASSED (Protocol validated) ▶ TEST 5: 1-Hour Ticket Reservation, Lockout & Expiry -> PASSED (Hold & auto-release verified) 🎉 ALL 5 SYSTEM TESTS PASSED WITH 100% INTEGRITY!
```

6.3 Development & Production Commands

```
# Install all dependencies across Turborepo npm install # Push database schema & seed initial Ethiopian raffles npm run db:push npm run db:seed # Start both Web Client (3000) and Admin Console (3001) npm run dev # Run automated verification suite ./node_modules/.bin/tsx scripts/test-platform.ts
```